Decomposed Nowcasting of Tropical Cyclone Satellite Cloud Imagery via Deterministic–Probabilistic Ensemble Learning

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***Abstract*—Accurate nowcasting of tropical cyclone satellite cloud imagery (TCSCI) is critical for disaster prevention and mitigation, but it remains challenging due to coupled uncertainties and multiscale dynamics in TCSCI evolution. While data-driven deep learning (DL)-based deterministic models mitigate biases inherent in traditional physics-driven approaches, their deterministic nature inherently limits uncertainty quantification. To address this, we propose TCSat-Diff, an ensemble framework that integrates spectral deterministic prediction with a frequency-adaptive probabilistic diffusion model. Specifically, we introduce a physics-inspired Reynolds decomposition strategy that separates TCSCI evolution into a large-scale trend flow and a small-scale fluctuating flow. The trend flow is predicted using the adaptive Fourier neural operator (AFNO) with mean-squared error (MSE) loss for smooth large-scale forecasts, while the fluctuating flow is captured by a conditional latent diffusion model (CLDM) for highly uncertain small-scale fluctuations. To further improve frequency-aware uncertainty modeling, we develop a signal-to-noise ratio (SNR)-guided noise frequency modulation (NFM) module and a time–frequency response (TiFR) module into the CLDM, enabling adaptive spectral focus during denoising. Experiments on 1099 tropical cyclones (TCs) from the Northwest Pacific demonstrate that TCSat-Diff outperforms previous deterministic and probabilistic baselines. Ablation studies confirm the effectiveness of the decomposition strategy and frequency-aware modules. Notably, cross-regional generalization evaluation on 480 TCs from the Southern Hemisphere reveals excellent generalization capability across different hemispheres. These results highlight the potential of TCSat-Diff for global TCSCI nowcasting, supporting early warning and risk reduction for TC hazards.**

***Index Terms*—Deterministic–probabilistic ensemble model, spatiotemporal prediction, tropical cyclone satellite cloud imagery (TCSCI).**

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I. Introduction

TROPICAL cyclones (TCs) are among the most destructive meteorological events, causing severe hazards and resulting in substantial loss of life and property damage [1], [2], [3], [4], [5]. With the advancement of satellite imaging technology, satellite observations have become an important tool for monitoring TCs [6], [7]. Researchers leverage TC satellite cloud imagery (TCSCI) to make accurate predictions about TC patterns, including TC intensity estimation [8], [9], [10], TC center location estimation [10], [11], [12], [13], TC wind radius estimation [14], [15], and so on [16]. These diverse applications highlight the importance of accurate TCSCI nowcasting for effective TC monitoring and disaster preparedness.

TCSCI nowcasting is a spatiotemporal sequence prediction task that uses historical cloud sequences to predict future ones. Research began in the 1970s, with early efforts focused on cloud tracking and motion estimation [17], [18]. Subsequent methods introduced cross correlation techniques for short-term cloud prediction [19]. However, traditional approaches mainly rely on cloud matching and linear extrapolation, thereby limiting their capacity to capture complex nonlinear dynamics, such as cloud deformation and rotation. As a result, their prediction performance is often inadequate for accurately modeling the complex spatiotemporal dynamics of atmospheric conditions.

Advancements in numerical weather prediction (NWP) techniques [20] have significantly enhanced the accuracy of cloud motion prediction. At the core of this approach lies the assimilation of multisource observational data to construct an initial atmospheric state that closely approximates reality. Based on this initial condition, a set of dynamical and thermodynamical equations governing atmospheric processes is then solved to simulate its spatiotemporal evolution [21]. However, inadequate understanding of atmospheric dynamics and data resolution constraints result in current physics-driven NWP models being incapable of fully explaining real-world dynamical processes [22], [23]. Compared to these models, data-driven deep learning (DL) models can automatically learn complex mapping relationships from data, potentially mitigating these limitations [24], [25], [26]. Commonly used architectures include ConvLSTM [27], PredRNN [28], [29], and others [30]. However, existing DL-based TCSCI predic-

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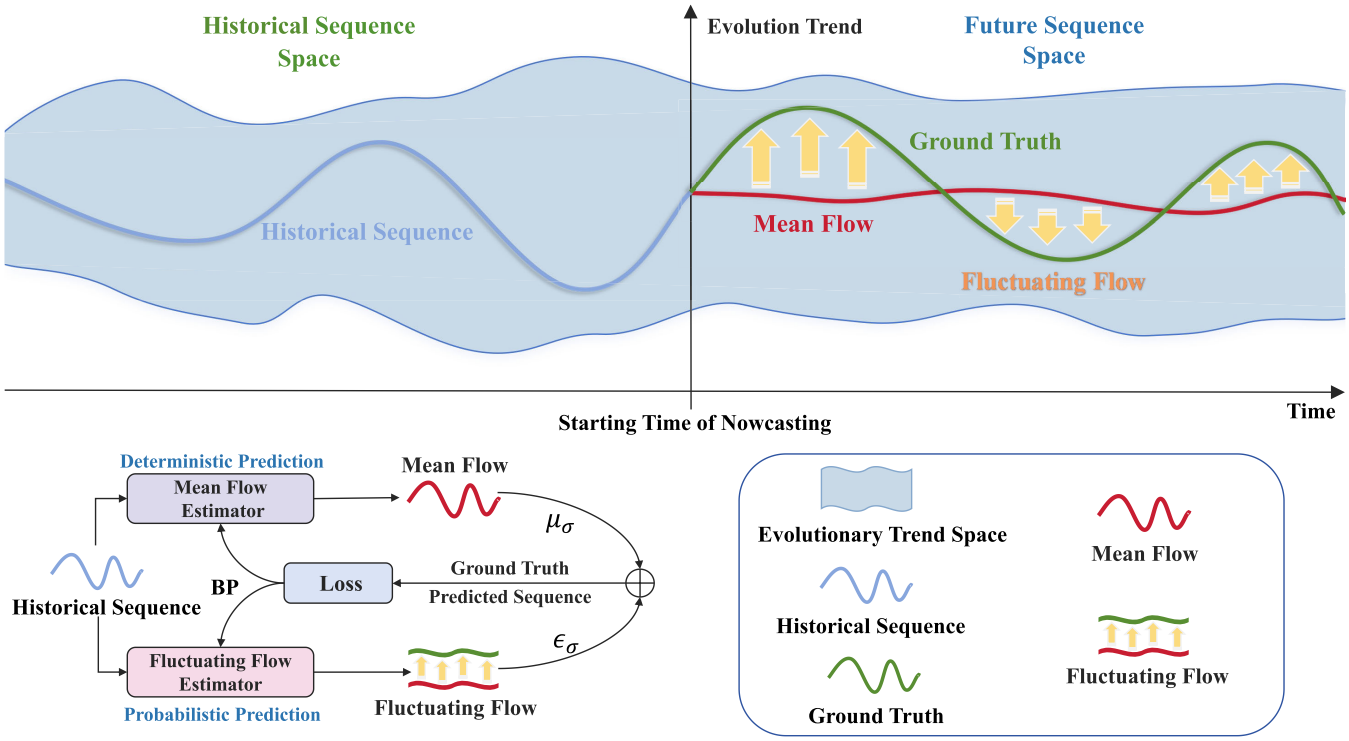


Fig. 1. Evolution trend of TCSCI is emulated as an irregular flow field with temporal fluctuations. The x-axis represents historical (negative) and predicted (positive) TCSCI sequences. The evolution is decomposed into two components: the mean flow and the fluctuating flow.

tion models are typically trained as deterministic models, optimized by minimizing mean-squared error (MSE) loss, which focuses on predicting average patterns [31]. From the perspective of grid resolution, such deterministic strategies inherently capture only the mean behavior of subgrid-scale motions. This often results in blurred predictions due to the loss of fine-scale details. Such models are inadequate for quantifying uncertainty in weather systems [32], potentially leading to prediction bias, particularly under the complex conditions of TCs.

As an alternative to deterministic modeling, probabilistic ensemble prediction uses multiple simulations sampled from the initial probability distribution to generate probable future atmospheric states [33], [34], [35]. However, it requires significant computational resources, limiting its practical applicability. Recent studies have explored diffusion models as an efficient alternative for probabilistic weather prediction [36], [37], [38]. Diffusion models consist of two core stages: the forward diffusion stage and the backward denoising stage. During the forward stage, Gaussian noise is incrementally added to the original data, gradually transforming the original data distribution to the standard Gaussian distribution. At the same time, a denoising model is trained to predict and remove the added noise based on the control conditions at each denoising time step, thereby learning the inverse mapping from the noise distribution to the original data distribution [39], [40]. In the backward stage, the trained denoising model leverages control conditions to map the noise distribution to the original distribution, generating corresponding samples [41], [42]. This iterative generation process makes diffusion models well-suited for modeling high-dimensional and complex data distributions. Consequently, the diffusion model presents a promising solution for modeling uncertainty in TCSCI nowcasting, particularly for capturing small-scale, high-frequency variability inherent in its evolution.

While recent advances in deterministic and probabilistic models have improved weather prediction, existing DL approaches are still limited in addressing the coupled dynamics and uncertainty inherent in TCSCI evolution. Due to the complexity of atmospheric conditions and limitations in current observational resolution, TCSCI evolution exhibits considerable randomness while maintaining order at the macroscale. This coupled nature of determinism and uncertainty can be analogized to irregular flow phenomena in fluid dynamics, as illustrated in Fig. 1. In fluid dynamics, Reynolds decomposition [43], [44] simplifies the modeling of turbulent flow by separating it into mean and fluctuating components, thereby distinguishing large-scale organized motion from small-scale fluctuations. Inspired by this, we decompose the TCSCI evolution into mean and fluctuating flows to simplify modeling. The mean flow is relatively stable, capturing the large-scale evolution of the TCSCI and representing the averaged effects of subgrid processes [45], [46], [47]. In contrast, the fluctuating component represents the variations in TCSCI evolution relative to the mean, accounting for the effects of unpredictable motion scales and inherently stochastic events at the current data resolution. These components are referred to as the mean flow and fluctuating flow.

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In this study, we propose **TCSat-Diff**, a deterministic–probabilistic ensemble model for the TCSCI nowcasting task. It consists of a deterministic part trained with MSE loss to estimate the mean flow, and a probabilistic part that estimates the fluctuating flow to correct the deterministic model's errors. The deterministic component employs the adaptive Fourier neural operator (AFNO), while the probabilistic component utilizes the conditional latent diffusion model (CLDM). The AFNO [48] is a spectral neural operator framework increasingly applied to model the dynamical evolution of atmospheric systems [49], [50]. It applies learnable spectral transformations in the Fourier domain, enabling efficient representation of large-scale dynamics and suppression of small-scale fluctuations [51]. This makes AFNO particularly well-suited for capturing the large-scale mean dynamics of TCSCI. The CLDM, based on the diffusion model framework, uses a stepwise reverse denoising process to sample from the complex distribution of fluctuating components. When conditioned on historical TCSCI sequences, it can generate fluctuating components consistent with the underlying conditional distribution. Notably, the denoising process in diffusion models is influenced by the signal-to-noise ratio (SNR): low-frequency structures are typically reconstructed early, while high-frequency details are recovered later. However, existing diffusion models lack explicit mechanisms to coordinate denoising across time–frequency scales, potentially leading to structural distortion and slow convergence. To address this, we propose two modules in CLDM: an SNR-guided noise frequency modulation (NFM) module and a time–frequency responsive (TiFR) module. These modules adapt the denoising dynamics to the spectral characteristics at each step in the backward denoising process. In summary, the key contributions of this work are shown as follows.

1) A deterministic–probabilistic ensemble model, named TCSat-Diff, is proposed for TCSCI nowcasting. By decoupling large-scale trend information and small-scale fluctuations, it simplifies the modeling of multiscale interactions and enables more focused, scale-aware predictions.
2) An SNR-guided NFM module and a time–frequency response (TiFR) module are introduced into CLDM, allowing the model to adjust its spectral focus across denoising time steps dynamically. This dynamic adaptation enhances structural reconstruction and improves the fidelity of fine-scale detail.
3) Extensive experiments on 1099 TCs from the Digital Typhoon dataset demonstrate that TCSat-Diff outperforms six deterministic and two probabilistic baselines across pixel-level, structure-level, and distribution-level evaluation metrics. Furthermore, a cross-regional generalization evaluation on 480 TCs from the Southern Hemisphere highlights the model's ability to generalize across different hemispheres.

II. Preliminaries and Problem Formulation

A. Adaptive Fourier Neural Operator

AFNO [48] is a frequency-domain neural operator designed to efficiently approximate solution operators for dynamical systems. Unlike conventional neural networks that map finite-dimensional vectors to other vectors, neural operators learn mappings from functions to functions [52], making them well-suited for modeling atmospheric dynamics with long-range dependencies [53]. By leveraging spectral convolution, AFNO captures global features and nonlocal interactions with high computational efficiency. As the backbone of our mean flow estimator, it offers three key advantages. First, it performs token mixing in the Fourier domain, effectively capturing low-frequency, large-scale evolution patterns. Second, its use of spectral sparsity, which discards high-frequency modes [54], introduces implicit smoothing that suppresses small-scale noise [55]. Finally, its architecture is computationally efficient, enabling a reduction in computational costs while maintaining excellent predictive performance.

B. Conditional Latent Diffusion Model

Diffusion models [56] are a class of probabilistic generative models inspired by nonequilibrium thermodynamics, which employ a parameterized Markov chain trained via variational inference. The CLDM [57] is a variant of the diffusion model. It employs a pretrained variational autoencoder (VAE) encoder [58] to map the original data into a lower-dimensional latent representation z_0 via the posterior distributional parameters (μ_{vae} and ϵ_{vae}), where μ_{vae} and ϵ_{vae} denote the mean and variance of the latent Gaussian distribution, respectively. The diffusion process is subsequently performed on this latent representation. In this process, the VAE primarily serves as a high-fidelity autoencoder that deterministically compresses and reconstructs the input imagery, rather than introducing strong stochastic behavior as in conventional VAE.

Specifically, given a time range $T > 0$ and a variance schedule $\beta_1, \ldots, \beta_T \in (0, 1)$, Gaussian noise is incrementally added during the forward diffusion process, generating a sequence of latent variables $\{z_t\}_{t=1}^{T}$. This process continues until the final latent variable z_T approximates a Gaussian distribution, as expressed below

$$q(z_t \mid z_{t-1}) = \mathcal{N}\left(z_t \mid \sqrt{1-\beta_t}\, z_{t-1}, \beta_t I\right) \tag{1}$$

where $t \in (1, 2, \ldots, T)$ denotes the tth time step in the time range T, $\mathcal{N}$ represents the Gaussian distribution, z_t is the latent variable at time step t, β_t is the noise increment parameter, and $\boldsymbol{I}$ is an identity matrix. The objective of the backward denoising process aims to progressively remove the noise from the noisy latent variable z_t, ultimately restoring the original latent representation

$$p_\theta(z_{t-1} \mid z_t, c) = \mathcal{N}(z_{t-1} \mid \mu_\theta(z_t, t, c), \Sigma_\theta(z_t, t, c)) \tag{2}$$

where c is the control condition (using the historical sequence in this study), which guides the denoising steps to ensure that the generated samples align with the desired objectives, as illustrated in Fig. 2. The $\mu_\theta(z_t, t, c)$ and $\Sigma_\theta(z_t, t, c)$ represent the estimated mean value and covariance matrix from z_t to z_{t-1} at time step t. At each backward denoising step, a denoising model predicts the noise and removes it, progressively reconstructing the denoised latent representation z_0. Once the denoising process is complete, a pretrained VAE decoder maps

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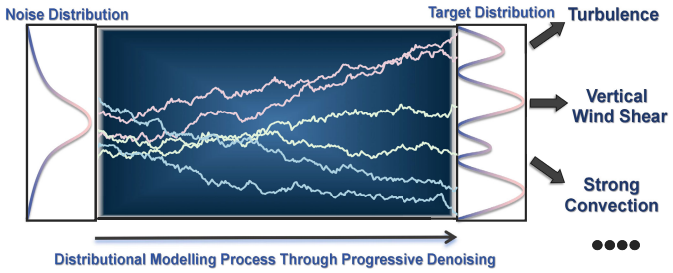


Fig. 2. Diffusion model progressively transforms samples from a standard Gaussian distribution (left) into a complex target distribution (right) through iterative denoising. (Middle) The denoising trajectories for different samples. The target distribution, which can be highly complex and nonlinear, represents phenomena such as turbulence, vertical wind shear, and strong convection, with each corresponding to distinct denoising trajectories.

z_0 back to the original space. During inference, only the backward denoising process is performed.

C. Denoising Frequency Tendency

The amplitude spectrum of natural images typically peaks at low frequencies and decays following a power law of $1/f^{\beta}$ [59]. Although infrared remote (IR) sensing images differ from natural visible-light images in imaging mechanisms and visual characteristics, they both exhibit spectra dominated by low-frequency components with sparse high-frequency components [60]. Moreover, IR images further emphasize this dominance of low-frequency information [61], [62], [63]. This low-frequency dominance is still noticeable, even after applying VAE compression [64]. In diffusion models, Gaussian noise with a uniform power spectrum is introduced at each denoising time step t. When added to the original image, this noise results in varying SNR across frequency bands [65]

$$\mathrm{SNR}(f) = \frac{E_{\mathrm{signal}}(f)}{E_{\mathrm{noise}}(f)} \tag{3}$$

where $E_{\mathrm{signal}}(f)$ and $E_{\mathrm{noise}}(f)$ represent the signal and noise energy at frequency f. In high-frequency bands where $E_{\mathrm{signal}}(f) < E_{\mathrm{noise}}(f)$, the SNR is low, and signal components are more easily masked by noise. Under the influence of SNR, the denoising model prioritizes low-frequency reconstruction in early denoising steps, restoring the global structure of the TCSCI. As the process continues, it progressively recovers higher-frequency components, refining fine-scale details in later stages [66], [67].

D. Problem Formulation

1) Problem Statement: The primary objective of this study is to predict the future sequence with τ frames, denoted as $y = \{f_{N+1}, f_{N+2}, \ldots, f_{N+\tau}\}$, conditioned on a historical TCSCI sequence with N frames, denoted as $x_p = \{f_1, f_2, \ldots, f_N\}$. To reduce computational costs, a pretrained VAE encoder maps the sequences x_p and y into the latent space representations $z_x = \{z_1, z_2, \ldots, z_N\}$ and $z_y = \{z_{N+1}, z_{N+2}, \ldots, z_{N+\tau}\}$, respectively (all sequences appearing below refer to sequences in latent space). The problem can be formally expressed as

$$\hat{z}_y = \mathrm{argmax}_{z_y} p\left(z_y \mid z_x\right) \tag{4}$$

where $\hat{z}_y = \left\{\hat{z}_{N+1}, \hat{z}_{N+2}, \ldots, \hat{z}_{N+\tau}\right\}$ denotes the predicted future sequence in latent space. Finally, the predicted latent sequence $\hat{z}_y$ is fed into the VAE decoder for reconstruction back to the original space to obtain the final prediction $\hat{y}$.

2) Task Decomposition: Given the historical TCSCI sequence z_x, we assume that all possible future trajectories $\tilde{z}_y$ lie within a trajectory space S. This space is analogized to an irregular flow field with temporal fluctuations. In fluid dynamics, Reynolds averaging is commonly used to simplify the turbulent flow modeling by decomposing fluid motion into mean and fluctuating parts

$$\phi = \bar{\phi} + \phi' \tag{5}$$

where ϕ denotes a solution variable in the Navier–Stokes equations, with $\bar{\phi}$ and ϕ' representing its mean and fluctuating components, respectively. The mean part $\bar{\phi}$ captures large-scale, deterministic trends by filtering out rapid fluctuations, while ϕ' represents fine-scale, uncertain turbulent motions. Similarly, we decompose the future latent sequence z_y into mean flow μ (the conditional expectation part) and fluctuating flow ϵ (the fluctuating part). In addition, this decomposition serves as a structural modeling design rather than a physical assumption of linear separability

$$z_y = \mu + \epsilon. \tag{6}$$

The μ is defined as the conditional expectation of z_y given z_x

$$\mu = \mathrm{E}\left[z_y \mid z_x\right] = \int \tilde{z}_y \cdot f\left(\tilde{z}_y \mid z_x\right) d\tilde{z}_y \tag{7}$$

where $f\left(\tilde{z}_y \mid z_x\right)$ denotes the conditional probability density function of $\tilde{z}_y$ given z_x. Conceptually, the μ represents the centroid trajectory within S, connecting the centers of all potential future evolutionary trajectories at each time step. The ϵ is defined as the residual between z_y and μ

$$\epsilon = z_y - \mathrm{E}\left[z_y \mid z_x\right]. \tag{8}$$

Given the complexity of TCSCI dynamics, accurately modeling μ and ϵ remains a significant challenge. Therefore, we employ neural networks to parametrically approximate them

$$\mu_\sigma = \sigma_m\left(z_x\right) \tag{9}$$

$$\epsilon_\sigma = \sigma_f\left(z_x\right) \tag{10}$$

$$\hat{z}_y = \mu_\sigma + \epsilon_\sigma \tag{11}$$

where μ_σ and ϵ_σ are the estimates of mean and fluctuating flows, respectively, parameterized by mean flow estimator σ_m and fluctuating flow estimator σ_f. Notably, the μ_σ and ϵ_σ are not equivalent to μ_{vae} and ϵ_{vae}. The μ_σ and ϵ_σ are task-specific features, whereas μ_{vae} and ϵ_{vae} are distributional parameters used to define the encoder's posterior.

III. Method

The architecture of TCSat-Diff is illustrated in Fig. 3. It comprises two complementary branches: a deterministic mean flow estimator and a probabilistic fluctuating flow estimator. Given historical satellite frames x_p, the encoder $\mathcal{E}$ first generates the latent representation z_x. Based on z_x, the mean flow

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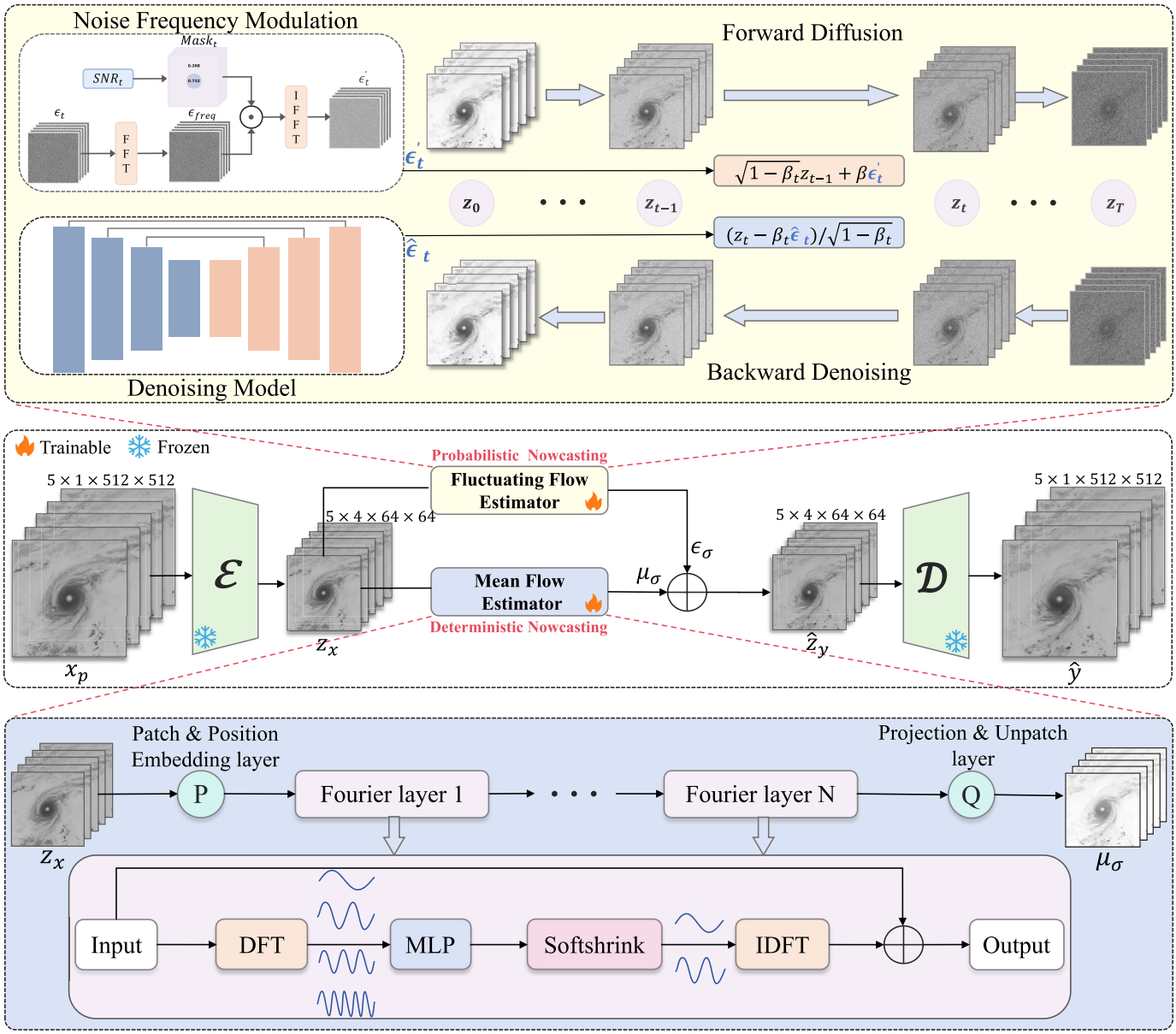


Fig. 3. Architecture of TCSat-Diff. The model integrates a deterministic mean flow estimator and a probabilistic fluctuating flow estimator. Given input frames x_p, features z_x are extracted by encoder $\mathcal{E}$. The predicted mean μ_σ and sampled fluctuating component ϵ_σ are combined and decoded by $\mathcal{D}$ to generate future predictions $\hat{y}$.

estimator predicts a deterministic mean μ_σ, capturing large-scale evolution patterns, while the fluctuating flow estimator models the uncertainty to generate a sample of the probabilistic fluctuating component ϵ_σ. The final prediction $\hat{y}$ is obtained by adding the estimated mean flow μ_σ and fluctuating component ϵ_σ, followed by decoding via the decoder $\mathcal{D}$ to produce a five-frame TCSCI sequence.

A. Deterministic Prediction Part

The mean flow estimator, based on an AFNO framework [48], captures large-scale evolution patterns in the TCSCI latent sequence, as illustrated in Fig. 3. The input sequence $z_x \in \mathbb{R}^{B\times F\times C\times H\times W}$ is partitioned into $N = (H/P) \times (W/P)$ nonoverlapping patches of size $P \times P$, where B, F, C, H, and W denote batch size, number of frames, number of channels, height, and width, respectively. Each patch is linearly projected and encoded to produce an embedding $z_{\text{emb}} \in \mathbb{R}^{(B\times F)\times N\times D}$

$$z_{\text{emb}} = \text{Proj}\left(\text{Patch}\left(z_x\right)\right) + PE \tag{12}$$

where Patch$(\cdot)$ and Proj$(\cdot)$ denote the patching and linear projection operations, respectively, and $PE \in \mathbb{R}^{(B\times F)\times N\times D}$ represents the positional encoding. The resulting embedding z_{emb} is then processed by a stack of AFNO layers. Within each layer, the embedding is first transformed into the frequency domain using a discrete Fourier transform (DFT)

$$z_{\text{freq}} = \text{DFT}\left(z_{\text{emb}}\right) \tag{13}$$

where z_{freq} denotes the frequency-domain representation. A frequencywise multilayer perceptron (MLP) is then applied

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to extract spectral features, followed by a soft-thresholding function S_λ to promote sparsity

$$\tilde{z} = \mathrm{S}_\lambda\left(\mathrm{MLP}\left(z_{\mathrm{freq}}\right)\right) \tag{14}$$

where $\mathrm{S}_\lambda(x) = \mathrm{sign}(x)\cdot\max(|x| - \lambda, 0)$ is the soft-thresholding function with sparsity parameter λ, promoting spectral sparsity to emphasize dominant low-frequency structures. The MLP consists of two linear layers with a GELU activation function, employing block-diagonal weights shared across patches [68]. An inverse DFT (IDFT) then maps the features back to the patch domain, followed by a residual connection. After N stacked AFNO layers, the output is projected and unpatched to reconstruct the spatial representation, generating the estimated mean flow μ_σ.

We optimize the parameters of the mean flow estimator using the MSE loss. In Section II-D, $\mu = \mathbb{E}\left[z_y \mid z_x\right]$ is defined as the conditional expectation of z_y given the historical sequence z_x. Assuming z_y given z_x follows a Gaussian distribution, denoted as $z_y \mid z_x \sim \mathcal{N}\left(\mu, \sigma^2\right)$, minimizing the MSE loss between the prediction $\sigma_m(z_x)$ and z_y is equivalent to maximum likelihood estimation (MLE) of the conditional mean μ [69], [70], [71], as expressed below

$$\begin{aligned}
&\mathrm{argmin}_{\sigma_m}\mathrm{MSE}\left(z_y, \sigma_m\left(z_x\right)\right) \\
&\Leftrightarrow \mathrm{argmin}_{\sigma_m} - \log\left(P\left(z_y \mid \mu, \sigma^2\right)\right) \\
&= \mathrm{argmin}_{\sigma_m} - \log\left(\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{\left(\mu - z_y\right)^2}{2\sigma^2}\right)\right) \\
&= \mathrm{argmin}_{\sigma_m}\frac{\left(\mu - z_y\right)^2}{2\sigma^2} + \log(\sigma\sqrt{2\pi}).
\end{aligned} \tag{15}$$

B. Probabilistic Prediction Part

The CLDM serves as the fluctuating flow estimator. As shown in Fig. 3, it incorporates two key components: an SNR-guided NFM module and a U-Net-based denoising model. The NFM module adjusts the spectral characteristics of the noise across denoising steps to enhance frequency-aware uncertainty modeling. To enable adaptive alignment with the time-varying spectral modulations introduced by the NFM module, a TiFR module is integrated into the denoising network, enhancing its responsiveness to spectral shifts.

1) SNR-Guided NFM Module: As shown in Section II-C, the denoising process exhibits frequency-dependent tendencies across steps due to SNR variations. However, standard diffusion models often lack explicit mechanisms to leverage this, potentially hindering the denoising efficiency and fidelity. To address this limitation, we introduce an SNR-guided NFM module. Given the noise $\epsilon_t \in \mathbb{R}^{B\times F\times C\times H\times W}$ at denoising step t, a 3-D fast Fourier transform (FFT) followed by spectral centering is applied along the temporal frames (F), height (H), and width (W) dimensions, transforming ϵ_t from the spatial to the frequency domain

$$\epsilon_{\mathrm{freq}} = \mathrm{FFT}\left(\epsilon_t\right) \tag{16}$$

where ϵ_{freq} is the complex-valued frequency-domain representation of ϵ_t, and FFT$(\cdot)$ denotes the combined 3-D FFT and spectral centering operation. A frequency mask is then calculated to modulate the amplitude of different frequency components

$$Mask_t = \begin{cases} 1 + \alpha_{\mathrm{LF}}\cdot\left(\dfrac{1}{1+SNR_t}\right), & \text{if } r \le (R_S\cdot r_{\max}) \\ \gamma + \alpha_{\mathrm{HF}}\cdot\left(\dfrac{SNR_t}{1+SNR_t}\right), & \text{if } r > (R_S\cdot r_{\max}) \end{cases} \tag{17}$$

$$SNR_t = \frac{\sqrt{1-\beta_t^2}}{\beta_t^2} \tag{18}$$

where $Mask_t$ is the frequency mask that modulates spectral components at step t, with α_{LF} and α_{HF} controlling the amplification of low and high frequencies, respectively. γ denotes the initial high-frequency noise intensity, SNR_t is the SNR value derived from β_t in (1), r denotes the distance from the spectral center, and R_S represents a ratio of the maximum radial distance $r_{\max}$ from the frequency center, determining the mask's frequency boundary between low- and high-frequency components. This design enables the model to prioritize low-frequency reconstruction during early denoising (when SNR is high) and gradually shift to high-frequency refinement as SNR decreases. The final mask is applied elementwise to ϵ_{freq}, followed by inverse FFT (IFFT) and inverse spectral centering to generate the frequency-modulated output

$$\epsilon_t' = \mathrm{IFFT}\left(\epsilon_{\mathrm{freq}} \times \mathrm{Mask}_t\right) \tag{19}$$

where ϵ_t' is the frequency-modulated noise used at diffusion step t, which is injected into the forward process of CLDM. Given the latent variable z_{t-1} at step $t-1$, the next latent variable z_t is sampled as

$$z_t = \sqrt{1-\beta_t}z_{t-1} + \sqrt{\beta_t}\epsilon_t'. \tag{20}$$

The objective of the backward denoising process is to reconstruct z_{t-1} from z_t. This is achieved by estimating the parameterized noise component $\hat{\epsilon}_t\left(z_t, t\right)$ using the denoising model, followed by:

$$z_{t-1} = \frac{1}{\sqrt{1-\beta_t}}\left(z_t - \sqrt{\beta_t}\hat{\epsilon}_t\left(z_t, t\right)\right). \tag{21}$$

Therefore, the training objective is to minimize the following loss function:

$$\mathcal{L}(\theta) = E_{t,z_0,\epsilon_t}\left[\left\|\epsilon_t' - \hat{\epsilon}_t\left(z_t, t\right)\right\|^2\right]. \tag{22}$$

Although the model is trained with frequency-scheduled noise, the reverse denoising process begins from standard Gaussian noise. At each denoising step, this module dynamically modulates the intensity of frequency-specific noise to enable the model to progressively focus on different spectral components. This aligns the denoising focus with the SNR characteristics at each denoising time step.

2) Denoising Model:

a) Model architecture: The denoising model aims to remove noise from the latent variable z_t by conditioning on historical context $z_x = z_1, z_2, \ldots, z_N$. To achieve this, the z_x and z_t are merged along the temporal dimension, forming a ten-frame input sequence for the denoising model. The denoising model is based on a 3D-UNet architecture, which effectively captures spatiotemporal contextual information by

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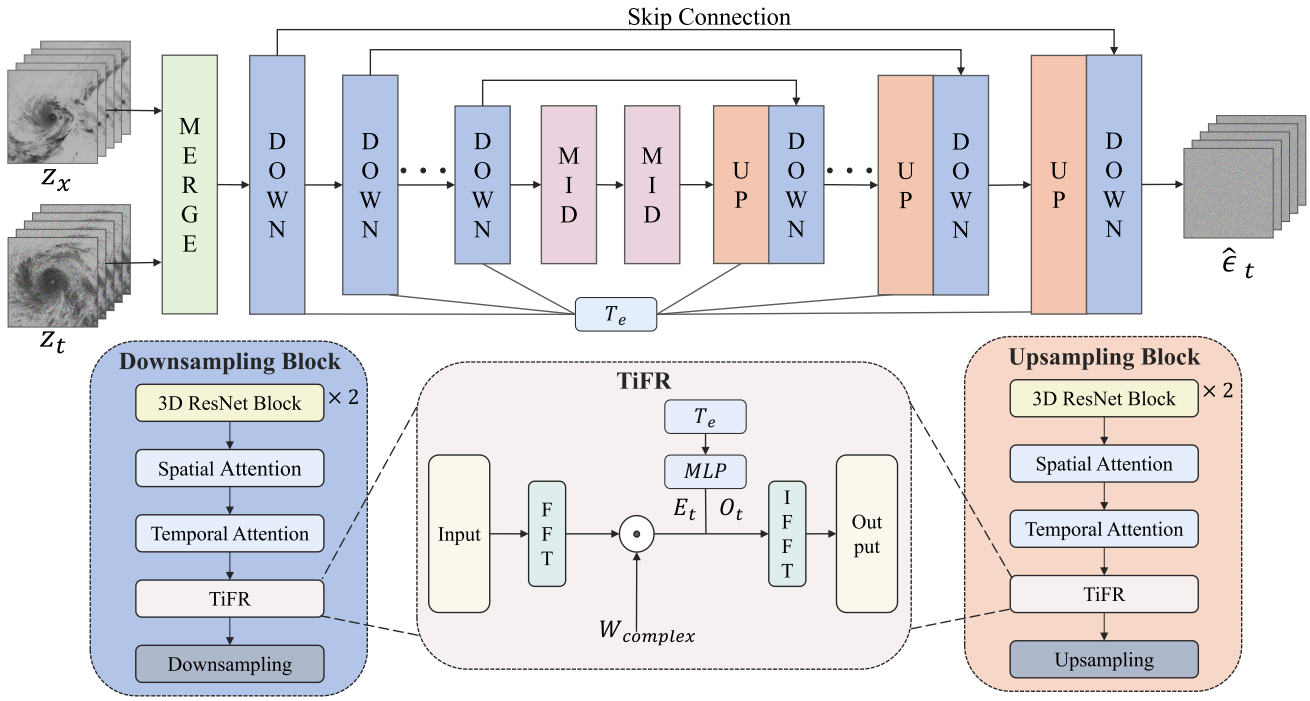


Fig. 4. Architecture of the CLDM's denoising network. The model adopts a U-Net-style encoder–decoder structure. Each downsampling and upsampling block incorporates stacked 3-D ResNet blocks, a spatial attention block, a temporal attention block, and a TiFR module to enhance spatiotemporal representation. The TiFR module is designed to help the model adapt to the specially constructed time-varying frequency noise.

aggregating both local and global features [72], [73]. As shown in Fig. 4, the network consists of a merging layer, four downsampling layers, two middle layers, and four upsampling layers. Each corresponding downsampling and upsampling layers are connected by skip connections. Each layer integrates multiple 3-D ResNet blocks, spatial and temporal attention blocks, and a TiFR module. In addition, 3-D convolution (or a 3-D transposed convolution) without padding is used for downsampling (or upsampling). The denoising time step t is embedded into a vector T_e using sinusoidal positional encoding [74]

$$T_e = \text{SinusoidalPosEmb}(t) \tag{23}$$

where SinusoidalPosEmb represents the operation of sinusoidal position encoding. This embedding T_e is provided to each block within the network.

b) TiFR module: To align the denoising model with the time-varying spectral noise characteristics introduced by the NFM module, we develop the TiFR module. This module enables adaptive spectral focusing at each denoising step.

Given an input feature map $X_f \in \mathbb{R}^{B\times F\times C\times H\times W}$ and the current time step $t \in \mathbb{R}^{B\times 1}$, a 3-D FFT is applied along the temporal (F), height (H), and width (W) dimensions, generating the complex frequency-domain representation X'_f. A learnable complex-valued weight matrix $W_{\text{complex}} \in \mathbb{R}^{B\times F\times C\times H\times W\times 2}$ is then used to dynamically modulate spectral components, with the final dimension representing real and imaginary parts

$$W'_{\text{complex}} = \text{Trans}\left(W_{\text{complex}}\right) \tag{24}$$

$$\widehat{X}_f = X'_f \odot W'_{\text{complex}} + \varepsilon \tag{25}$$

where Trans$(\cdot)$ denotes the merging of real and imaginary parts into complex form, and ε is a small constant added for numerical stability. The denoising time step embedding T_e is then mapped by an MLP into denoising time scale $E_t \in \mathbb{R}^{B\times C}$ and denoising time offset $O_t \in \mathbb{R}^{B\times C}$

$$E_t, O_t = \text{Chunk}(\text{MLP}(T_e)) \tag{26}$$

where Chunk$(\cdot)$ denotes average splitting operation. The E_t and O_t modulate $\widehat{X}_f$ to dynamically adapt frequency features across denoising time steps

$$X_{\text{Weighted}} = \widehat{X}_F \odot (1 + E_t) + O_t. \tag{27}$$

Finally, the output $O_f \in \mathbb{R}^{B\times F\times C\times H\times W}$ is obtained by transforming X_{Weighted} back to the spatiotemporal domain using the inverse FFT

$$O_f = \text{IFFT}\left(X_{\text{Weighted}}\right). \tag{28}$$

This module enables the model to adaptively respond to the frequency-specific noise targets modulated by the NFM at each denoising time step.

IV. Experiment

A. Dataset and Implementation Details

We use the Digital Typhoon Dataset [75] as the primary benchmark. It contains 189 364 IR satellite images (11-μm channel) of 1099 TCs recorded by the Himawari satellites over the Northwest Pacific (100°E–180°E, Northern Hemisphere) from 1978 to 2022. Each grayscale image covers a 2500 × 2500-km region centered on the TC, with 5-km spatial resolution and 1-h temporal resolution, and is projected to 512 × 512 pixels using an equal-area azimuthal projection.

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We group every ten frames into a temporal sequence, yielding 178 325 TCSCI sequences. The dataset is split into 60% training (106 995), 20% validation (35 665), and 20% test (35 665) samples. For each sequence, the first five frames serve as the input and the latter five as the ground truth.

To evaluate cross-regional generalization, we use an independent benchmark from the Southern Hemisphere [76], an extension of the Digital Typhoon dataset. Spanning 90°E–160°E from 1979 to 2024, this dataset comprises 480 TCs. Notably, this dataset is not used during training, providing a cross-regional generalization evaluation of the model's ability to generalize across hemispheres.

Following the original dataset preprocessing protocol, pixel values are clipped to [150, 350] and then normalized to [–1, 1]. For the diffusion model, we employ the original denoising diffusion probabilistic models (DDPMs) [77] sampler with 100 sampling steps. In (17), the hyperparameters $\gamma = 0.7$, $R_S = 0.2$, $\alpha_{\mathrm{LF}} = 0.3$, and $\alpha_{\mathrm{HF}} = 0.6$ are selected through hyperparameter tuning on the validation set. All experiments are conducted using PyTorch on eight NVIDIA Tesla V100 GPUs (each with 32-GB memory).

B. Evaluation Metrics

Seven metrics are employed to comprehensively evaluate our model: mean absolute error (MAE), MSE, mean bias error (MBE), Pearson correlation coefficient (PCC), structural similarity index measure (SSIM) [78], peak SNR (PSNR) [79], and Fréchet video distance (FVD) [80]. Pixel-level accuracy is assessed using MAE, MSE, and MBE, while SSIM and PSNR quantify structural fidelity and perceptual quality. PCC captures the statistical correlation between predicted and ground truth sequences. FVD evaluates distribution-level similarity in a learned feature space, with lower values indicating improved realism, temporal coherence, and diversity. Definitions for each metric are detailed below

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|x_i - y_i| \tag{29}$$

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(x_i - y_i)^2 \tag{30}$$

$$\mathrm{MBE} = \frac{1}{N}\sum_{i=1}^{N}(x_i - y_i) \tag{31}$$

$$\mathrm{PCC} = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{N}(x_i - \bar{x})^2}\cdot\sqrt{\sum_{i=1}^{N}(y_i - \bar{y})^2}} \tag{32}$$

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{\mathrm{xy}} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{33}$$

$$\mathrm{PSNR} = 10\cdot\log_{10}\left(\frac{\mathrm{peakval}^2}{\mathrm{MSE}}\right) \tag{34}$$

$$\mathrm{FVD} = \left\|\mu_r - \mu_g\right\|^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r\Sigma_g\right)^{\frac{1}{2}}\right) \tag{35}$$

where N is the number of samples in the test set; x_i and y_i denote the predicted and ground truth sample of the ith, respectively; $\bar{x}$ and $\bar{y}$ are the means of x and y, respectively; μ_x and μ_y are local means; σ_x^2 and σ_y^2 are local variances; σ_{xy} is the local covariance; C_1 and C_2 are stabilization constants in SSIM; peakval is the maximum value in the data; and μ_r and μ_g and Σ_r and Σ_g are the means and covariances of the real and generated feature distributions (extracted by a pretrained StyleGAN-v network [81]), respectively.

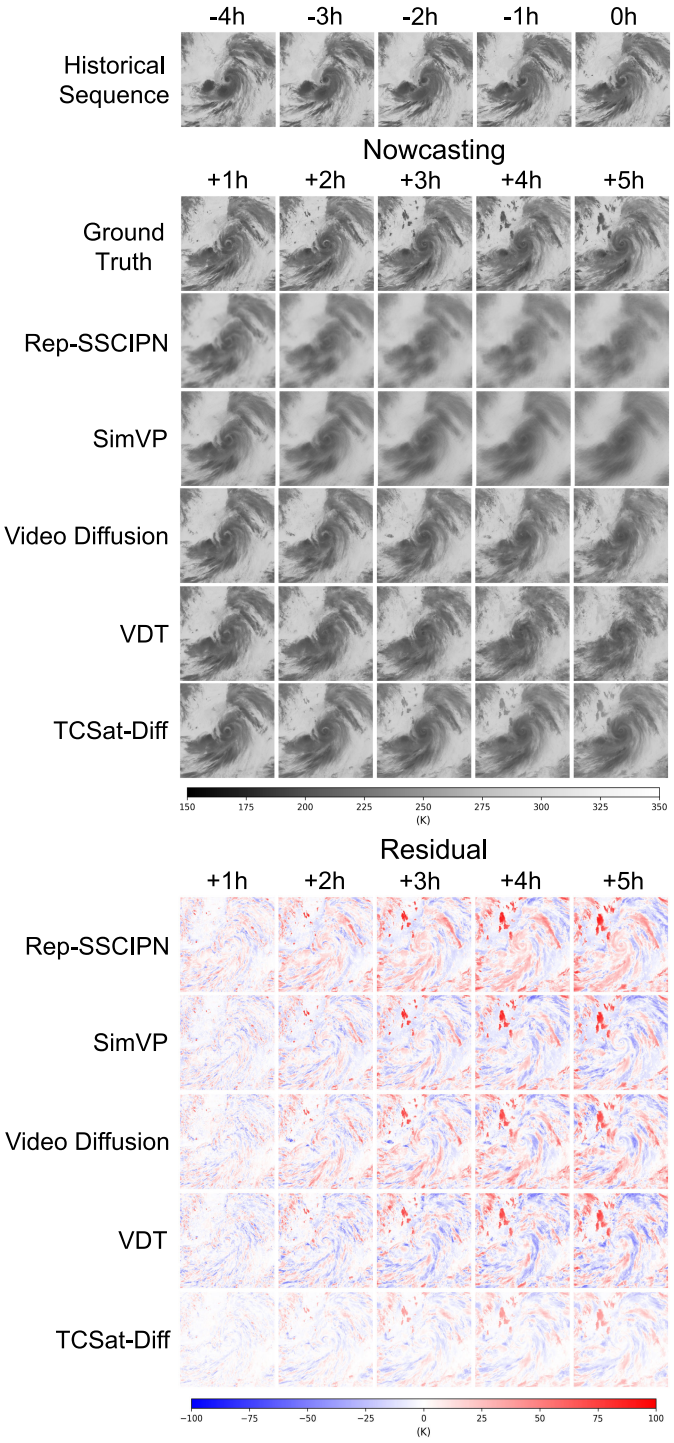


Fig. 5. Visual analysis of Super TC Doksuri (2023) in the Northern Hemisphere, using TCSCI inputs from 02:00 to 06:00 UTC on July 27, 2023, to predict the evolution from 07:00 to 11:00 UTC. TCSat-Diff produces predictions that closely align with observed cloud structures and exhibit substantially lower residuals, indicating improved accuracy in capturing both large-scale structure and fine-scale details.

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TABLE I

QUANTITATIVE COMPARISON OF OVERALL PERFORMANCE ON THE TEST SET. THE BEST VALUE IS HIGHLIGHTED IN RED, WHILE THE SECOND BEST IS HIGHLIGHTED IN BLUE

Method	MAE↓	MSE↓	MBE↓	PCC↑	PSNR (dB)↑	SSIM↑	FVD↓	Params (M)↓	Running Time (s)↓
Persistence	13.066	399.864	-0.45	0.659	20.528	0.587	1996.781	-	-
MAU	9.433	217.962	2.369	0.794	23.106	0.721	1573.876	70.541	0.462
PredRNN	8.882	204.172	-0.220	0.866	23.955	0.720	1430.254	131.882	0.767
SwinLSTM	9.537	229.082	3.674	0.813	23.115	0.730	1601.777	80.643	0.585
TAU	8.658	160.678	-0.031	0.853	24.388	0.713	1773.631	75.996	0.659
Rep-SSCIPN	8.944	178.648	0.375	0.832	23.996	0.726	1547.518	67.405	0.576
SimVP	8.452	155.679	0.257	0.879	24.564	0.724	1321.598	79.314	0.543
Video Diffusion	8.056	159.294	-0.703	0.867	24.411	0.724	590.962	140.786	5.149
VDT	8.558	188.353	-0.873	0.850	23.931	0.714	772.268	433.389	13.125
TCSat-Diff (DDPM 100)	7.887	142.044	-0.484	0.880	25.055	0.763	550.404	105.238	4.742
TCSat-Diff (DDIM 25)	7.992	148.224	-0.492	0.876	24.836	0.756	576.891	105.238	1.125
TCSat-Diff (PLMS 25)	7.916	145.124	-0.498	0.876	24.921	0.757	561.125	105.238	0.958
TCSat-Diff (DPM-Solver 25)	7.898	143.981	-0.516	0.878	24.981	0.761	556.981	105.238	0.791

C. Experiment Results

For comparison with existing methods, we select six deterministic baselines: MAU [82], PredRNN [83], SwinLSTM [84], TAU [85], Rep-SSCIPN [30], SimVP [86], and two probabilistic baselines: video diffusion [87] and VDT [88]. As a persistence baseline, the last input frame is replicated five times to generate the persistence prediction, serving as a reference to evaluate the model's ability to capture dynamic changes.

The evaluation metrics described in Section IV-B are used to quantitatively compare the different methods, as presented in Table I. The results show that the proposed TCSat-Diff outperforms other methods across most metrics. Notably, diffusion-based methods perform better than deterministic models in terms of FVD. Compared to the representative deterministic model SimVP, video diffusion achieves a 55.3% reduction in FVD (from 1321.598 to 590.962), VDT achieves a 41.6% reduction in FVD (from 1321.598 to 772.268), and TCSat-Diff achieves a 58.4% reduction in FVD (from 1321.598 to 550.404). This is because deterministic models tend to collapse toward mean predictions, limiting their ability to capture the diversity of the target distribution. This leads to reduced visual fidelity and sample diversity. In contrast, diffusion models alleviate this limitation. However, they underperform on pixel-level metrics, such as MSE and PSNR, since they optimize the Kullback–Leibler divergence rather than MSE loss. By integrating both deterministic and probabilistic modeling paradigms, TCSat-Diff achieves a well-balanced performance in these metrics. Specifically, it reduces MAE and MSE by an average of 15.2% and 32.5%, respectively, while improving PCC by 6.8% compared to the baseline models. Furthermore, it reduces FVD by 65.7% compared to deterministic methods and by 19.3% compared to other diffusion-based models. These results demonstrate that TCSat-Diff offers consistent improvements across pixel-level, structure-level, and distribution-level metrics.

Parameter count and inference time quantify the computational cost of each model. Among diffusion-based models, TCSat-Diff has the fewest parameters (105.238 M) and the fastest inference time (4.742 s), while achieving superior performance across multiple evaluation indicators. However, diffusion-based models inherently require longer inference time than deterministic models due to their iterative sampling process. Given their advantage in modeling uncertainty, this computational cost is well justified. We further experimented by replacing the original DDPM 100-step sampling with 25-step sampling using the denoising diffusion implicit models (DDIMs) [89], pseudolinear multistep (PLMS) method [90], and DPM-Solver [91] samplers. The experimental results show that the lighter samplers (25 steps) significantly reduce inference time with minor loss of accuracy, especially when using the DPM-Solver sampler. These results demonstrate that TCSat-Diff achieves a good balance between precision and efficiency, with potential for further acceleration.

D. Case Study on Super TC Doksuri

To assess the performance of different models on TC cases, we use Super TC Doksuri (2023) as a case study. Doksuri was the most intense and destructive TC that made landfall in China in 2023, ranking among the top ten natural disasters of the year [92]. It affected approximately 2.95 million people and caused direct economic losses of RMB 14.95 billion [93]. We compare a persistence baseline with two deterministic models (Rep-SSCIPN and SimVP) and two probabilistic generative models (video diffusion and VDT). The quantitative performance of each model throughout Doksuri's lifecycle is summarized in Table II.

Quantitative experimental results demonstrate that while diffusion-based methods achieve lower FVD values, they underperform on MSE, MBE, PCC, and PSNR compared to the deterministic model SimVP. In contrast, the proposed TCSat-Diff achieves notable improvements in most metrics. Compared to the baseline models, TCSat-Diff achieves a 24.4% reduction in MAE and a 41.1% reduction in MSE on average, with only a slight compromise in MBE. It further improves PCC, PSNR, and SSIM by 9.2%, 8.7%, and 11.9%, respectively, while reducing FVD by 60.1%. This result demonstrates the superior capability of TCSat-Diff in handling extreme Super TC events.

To intuitively demonstrate the differences in model predictions, we visualize the evolution of TCSCI from 07:00 to 11:00 UTC on July 27, 2023, using observational data from 02:00 to 06:00 UTC as input. During this forecast window, Doksuri intensified from a strong TC to a Super TC

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TABLE II
PERFORMANCE ON SUPER TC DOKSURI (2023)

Method	MAE↓	MSE↓	MBE↓	PCC↑	PSNR (dB)↑	SSIM↑	FVD↓
Persistence	15.698	502.498	0.089	0.634	19.056	0.503	1996.407
Rep-SSCIPN	10.407	211.219	0.906	0.833	22.838	0.663	2053.636
SimVP	9.769	184.142	-0.472	0.858	23.421	0.681	1819.860
Video Diffusion	9.488	193.822	0.529	0.854	23.196	0.678	627.675
VDT	9.874	207.324	-1.014	0.845	22.914	0.683	680.089
TCSat-Diff (ours)	8.357	153.035	-0.732	0.887	24.223	0.729	573.298

TABLE III
ABLATION ANALYSIS OF TCSAT-DIFF WITH DIFFERENT COMPONENTS, RESPECTIVELY

Mean Flow Estimator	Fluctuating Flow Estimator			MAE↓	MSE↓	PSNR (dB)↑	SSIM↑	FVD↓	Params↓	Running time↓
	+DDPM	+NFM	+TiFR							
✓				9.012	191.548	23.631	0.721	1432.872	21.442	0.270
✓	✓			8.028	160.515	24.676	0.751	632.751	98.318	4.545
✓	✓	✓		7.921	149.613	24.691	0.754	592.142	98.318	4.545
✓	✓	✓	✓	7.887	142.044	25.055	0.763	550.404	105.238	4.742

(09:00 UTC), providing an excellent scenario to assess the ability of each model to capture rapid intensification. As illustrated in Fig. 5, deterministic methods exhibit limited capability in capturing structural details, particularly in the accurate reconstruction of TC edge structures and fine-grained details. Although video diffusion and VDT alleviate these limitations, their residual maps still exhibit substantial prediction errors that intensify over time. In contrast, TCSat-Diff effectively reconstructs structural features and fine-grained details of the TCs while maintaining low prediction bias. Moreover, as lead time increases, both video diffusion and VDT exhibit increasing structural degradation in the predictions, particularly in the eyewall. In contrast, TCSat-Diff demonstrates improved structural coherence, effectively maintaining the integrity of cyclone features. Both quantitative and qualitative analyses confirm that TCSat-Diff exhibits excellent robustness under Super TC conditions.

E. Ablation Study

We perform ablation experiments to confirm the effectiveness of the proposed modules. We conduct experiments under four configurations. The first group does not decouple the tasks and uses AFNO trained with MSE loss for prediction. The second group decomposes the task by predicting the mean flow estimator using AFNO and fluctuating flow using CLDM sampled by DDPM, and then sums both parts to obtain the final prediction. The third group builds on the second group by adding an NFM module to the CLDM. The fourth group builds upon the third group by adding TiFR modules to the CLDM. The results from the test set are shown in Table III.

We first verify the effectiveness of the task decomposition. As shown in Table III, the first group underperforms compared to other groups across all metrics. Compared to using AFNO alone, incorporating DDPM leads to a 10.9% reduction in MAE (from 9.012 to 8.028) and a 16.2% reduction in MSE (from 191.548 to 160.515). PSNR improves by 4.4% (from 23.631 to 24.676 dB), SSIM increases by 4.1% (from 0.721 to 0.751), and the FVD decreases significantly by 55.8% (from

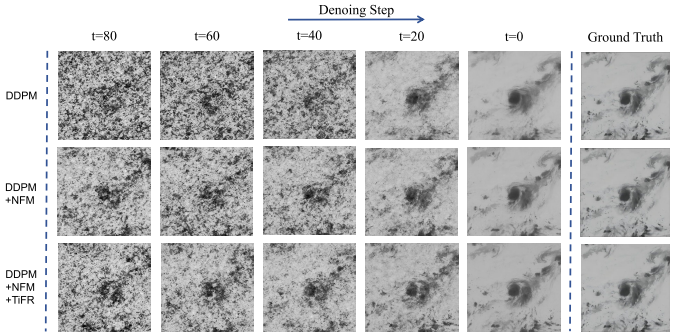


Fig. 6. Visualization of the denoising process for fluctuating flow estimator component ablation experiments. To clearly show the difference in the final effect of different groups, the results for all groups are shown after adding the mean flow predictions.

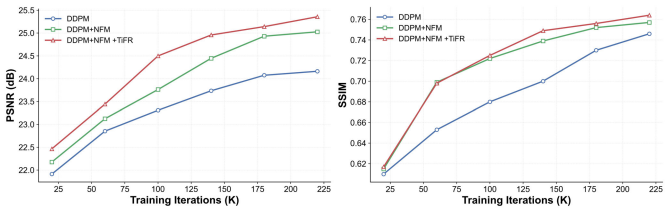


Fig. 7. Variation curves of different PSNR (left) and SSIM (right) of the training process for fluctuating flow estimator components ablation experiments.

1432.872 to 632.751). This is mainly because the evolution of TC morphology is influenced by complex environmental factors, leading to inherent randomness and uncertainty. It is difficult for deterministic models to identify and generalize universal patterns applicable to all TCs from such coupled evolutionary patterns. In contrast, the task decomposition strategy effectively simplifies the task and gets better results. However, it also introduces a certain computational cost. When DDPM is removed, the AFNO-only group achieves the lowest parameter count (21.442 M) and inference time (0.270 s) across all groups. Considering its substantial performance gains, the computational cost introduced by DDPM is well justified. We next verify the importance of the NFM module in

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TABLE IV
PERFORMANCE COMPARISON UNDER DIFFERENT ENSEMBLE STRATEGIES

Ensemble Strategy	MAE↓	MSE↓	MBE↓	PCC↑	PSNR (dB)↑	SSIM↑	FVD↓	CRPS↓
Single-sample	7.887	142.044	-0.484	0.880	25.055	0.763	550.404	0.041
3-member mean	7.523	137.574	-0.417	0.881	25.238	0.746	750.553	0.030
5-member mean	7.270	126.587	-1.140	0.890	25.593	0.765	847.839	0.028
10-member mean	6.914	111.398	-0.730	0.901	26.029	0.780	974.185	0.025
20-member mean	6.894	109.413	-0.735	0.903	26.094	0.784	1023.411	0.024
30-member mean	6.864	107.470	-0.405	0.907	26.191	0.789	1060.967	0.024
50-member mean	6.835	107.030	-1.245	0.909	26.223	0.795	1079.873	0.024

this task. As shown in Table III, the NFM module significantly improves the performance of the model. Notably, the NFM module, active only during training, introduces no additional inference cost. To directly demonstrate the impact of the NFM module, we visualize the denoising process for each group that includes a diffusion model. As shown in Fig. 6 , the standard DDPM model slowly recovers the coherent TC structure from the noise. The distinctive spiral structure only emerges midway through the denoising process. This is because, although standard denoising models prioritize low-frequency structures in the early stages, they still attempt to recover high-frequency details due to high-frequency noise. However, in the early denoising stages with high noise content, recovering high-frequency components is challenging due to their sensitivity to noise. The large prediction weight in this stage (due to the high noise variance from the forward diffusion process) amplifies errors in high-frequency details. This leads to biases that are difficult to correct later and ultimately degrade the recovered image structure. With the introduction of the NFM module, the model focuses more on recovering low-frequency structure in the early stage, enabling it to quickly recover the spiral structure of the TC in the early stage of denoising. In the later stages, it shifts its focus to recovering high-frequency details, thereby preserving fine-grained structures and enhancing realism.

Finally, we verify the contribution of the TiFR module. As shown in Fig. 6 and Table III, the group with the TiFR module performs better than the group without TiFR on all indices, and it demonstrates a closer match to real observations in the visual results, especially in the detail restoration. We also examine the PSNR and SSIM changes during training in the different experimental groups, as shown in Fig. 7 . Experimental results indicate that the group with the TiFR module exhibits faster convergence, improved early-stage prediction, and superior overall accuracy. This suggests that the TiFR module can help the model better adapt to different noise frequencies at different time steps and achieve better results. Meanwhile, the TiFR module has minimal computational cost, validating its efficient design.

F. Analysis of Ensemble Strategies

To investigate the impact of ensemble strategies on model performance and uncertainty reduction, we conduct a systematic evaluation of TCSat-Diff under different ensemble strategies. Specifically, we compare single-sample inference with ensemble means computed over 3, 5, 10, 20, 30, and 50 ensemble members. The continuous ranked probability score (CRPS) [94] is additionally introduced to measure the performance of ensemble prediction. CRPS is a standard metric for evaluating probabilistic predictions, measuring how well the predicted marginal distributions align with the ground truth [31]. In expectation, it is minimized when the prediction accurately reflects the true predictive uncertainty. The calculation is defined as follows:

$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{+\infty} (F(c) - H(c - y))^2 \, \mathrm{dc} \tag{36}$$

where $F(c)$ denotes the cumulative distribution function (cdf) of the predicted variable, y is the ground-truth observation, and $H(c-y)$ is the Heaviside function. As shown in Table IV, increasing the number of ensemble members consistently improves performance across multiple evaluation metrics, including MAE, MSE, PCC, PSNR, and SSIM. However, FVD exhibits an upward trend with larger ensemble sizes. This phenomenon arises because, while ensemble averaging improves statistical consistency, it also tends to lead to overly deterministic and smoothed predictions, suppressing prediction diversity. This increases the distributional distance from the ground truth in feature space. Moreover, when the ensemble size exceeds 20 members, further performance improvement from ensemble averaging becomes marginal. Consequently, while ensemble methods provide superior numerical performance, their effect on perceptual quality requires careful balance and further optimization.

G. Cross-Regional Generalization Experiment on Southern Hemisphere

The TCSat-Diff model is trained exclusively on TCSCI from the Northern Hemisphere. To evaluate its cross-hemispheric generalization capability, we evaluate it on the Southern Hemisphere dataset without any fine-tuning. As shown in Table V, TCSat-Diff achieves the best performance across most evaluation metrics, including a substantial reduction in MAE (8.995) and MSE (196.393), and the highest scores in PCC (0.837), PSNR (23.861 dB), and SSIM (0.751). Notably, it also obtains the lowest FVD (642.218), indicating excellent perceptual quality and distribution alignment with the ground truth. Compared to both deterministic and diffusion-based baselines, TCSat-Diff consistently demonstrates enhanced numerical accuracy and structural fidelity, without requiring additional training. These results demonstrate the model's excellent cross-regional generalization capability, enabling reliable nowcasting of TC dynamics across hemispheric regimes.

We select TC Kirrily (2024), located in the Southern Hemisphere, as a case study for further qualitative analysis.

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TABLE V
QUANTITATIVE COMPARISON OF CROSS-REGIONAL GENERALIZATION EXPERIMENT

Method	MAE↓	MSE↓	MBE↓	PCC↑	PSNR (dB)↑	SSIM↑	FVD↓
Persistence	12.503	407.830	-0.194	0.653	20.735	0.580	2294.906
Rep-SSCIPN	10.425	254.458	0.022	0.810	22.978	0.701	1934.310
SimVP	10.681	255.372	-0.312	0.797	22.908	0.623	1797.925
Video Diffusion	9.720	233.814	-0.660	0.815	23.239	0.715	677.819
VDT	9.841	234.370	-0.857	0.808	23.127	0.722	782.467
TCSat-Diff (ours)	8.995	196.393	-0.462	0.837	23.861	0.751	642.218

As illustrated in Fig. 8, all methods exhibit varying degrees of structural distortion in this region compared to results from the Northwest Pacific. Notably, diffusion-based models fail to capture the structural evolution of the cyclone, with errors becoming increasingly pronounced over time. This can be attributed to the Coriolis effect, which induces opposite rotational directions for TCs across hemispheres, thus introducing fundamental dynamic disparities that complicate cross-hemispheric prediction tasks. In contrast, TCSat-Diff effectively leverages the historical observational frames to infer the underlying evolution of TCSCI, preserving the structural coherence and fine-scale details of the cyclone. These results demonstrate the superior generalization ability and robustness of TCSat-Diff in modeling Southern Hemisphere cyclones without requiring additional retraining.

H. Hyperparameters Sensitivity Analysis

The hyperparameters of the NFM module determine the balance between low- and high-frequency modulation. Appropriate tuning of these hyperparameters is essential for ensuring model stability and reconstruction fidelity. To quantify their impact, we perform a comprehensive sensitivity analysis on four key hyperparameters: γ, R_S, α_{LF}, and α_{HF}, as defined in (17). The sensitivity ranges of these parameters are determined based on theoretical considerations to ensure training stability and prevent over-amplification of spectral components.

We first analyze γ, which modulates the initial intensity of high-frequency components. Its value varies from 0.6 to 1.1 to assess the influence of high-frequency amplification on model performance. As shown in Fig. 9(a), increasing γ across this range leads to a reduction in PCC and an increase in FVD. This performance degradation occurs because a higher γ excessively amplifies high-frequency signals in the early denoising stages, which contradicts the inherent low-frequency prioritization of diffusion models. This premature activation disrupts the coarse-to-fine denoising process, leading to unstable reconstructions. Conversely, an excessively small γ (e.g., $\gamma = 0.6$) maintains a high PCC but elevates FVD compared to $\gamma = 0.7$. This indicates that excessively small γ offers limited gains in statistical accuracy at the expense of perceptual fidelity. A moderate value of $\gamma = 0.7$ provides the best tradeoff between statistical accuracy and fidelity.

The hyperparameter R_S defines the frequency boundary of the mask as a proportion of the maximum radial distance from the spectral center, thereby controlling the transition between low- and high-frequency modulation. We assess its impact by varying R_S from 0.1 to 0.6. The results, shown in Fig. 9(b), demonstrate that as R_S increases, PCC decreases slightly while FVD increases. Smaller values of R_S prioritize the recovery of global structures, whereas larger values prematurely introduce high-frequency details. A suitable setting of $R_S = 0.2$ achieves the most stable reconstruction, which provides the most stable reconstruction by balancing statistical accuracy (PCC) with perceptual fidelity (FVD).

We next investigate α_{LF}, which controls the amplification strength of low-frequency modulation. Varying α_{LF} from -0.1 to 0.4 in Fig. 9(c) reveals that increasing its value steadily improves PCC while slightly reducing FVD. Higher values of α_{LF} amplify the mask's low-frequency response, thereby enabling the model to more effectively capture coarse structural priors in the early denoising stages. However, once α_{LF} exceeds 0.3, the performance gains saturate because excessive low-frequency dominance inhibits the learning of fine-scale details. Therefore, $\alpha_{\mathrm{LF}} = 0.3$ is selected as the optimal setting, delivering the most consistent reconstruction quality with well-preserved structural integrity.

Finally, we analyze α_{HF}, which governs the enhancement of high-frequency components. Its effect is tested across the range of 0.3–0.8. As shown in Fig. 9(d), increasing α_{HF} initially raises PCC before a slight decline, while FVD exhibits an inverse trend. This reveals that moderate enhancement of high-frequency components refines details and improves fidelity. Conversely, excessive amplification degrades overall performance. Consequently, $\alpha_{\mathrm{HF}} = 0.6$ is identified as the value that yields the most stable reconstruction.

V. CONCLUSION

We propose TCSat-Diff, a deterministic–probabilistic ensemble model for TCSCI nowcasting. The model captures large-scale mean evolution via an AFNO-based deterministic branch optimized with MSE loss and models inherent uncertainty components through a CLDM. To enhance frequency-aware uncertainty modeling, the CLDM is equipped with a time-varying NFM module and a time–frequency response mechanism, enabling dynamic adaptation across spectral bands.

Extensive experiments on 1099 Northern Hemisphere TCs demonstrate that TCSat-Diff achieves accurate predictions while preserving fine-grained structural detail. A case study on Super TC Doksuri (2023) highlights its robustness, capturing critical features with minimal bias despite the cyclone's complex and rapidly evolving dynamics.

Ablation studies validate the importance of integrating deterministic and probabilistic strategies, as well as the contributions of the NFM and TiFR modules. Ensemble analysis reveals that increasing the number of ensemble members leads to better numerical performance but increases the FVD. We

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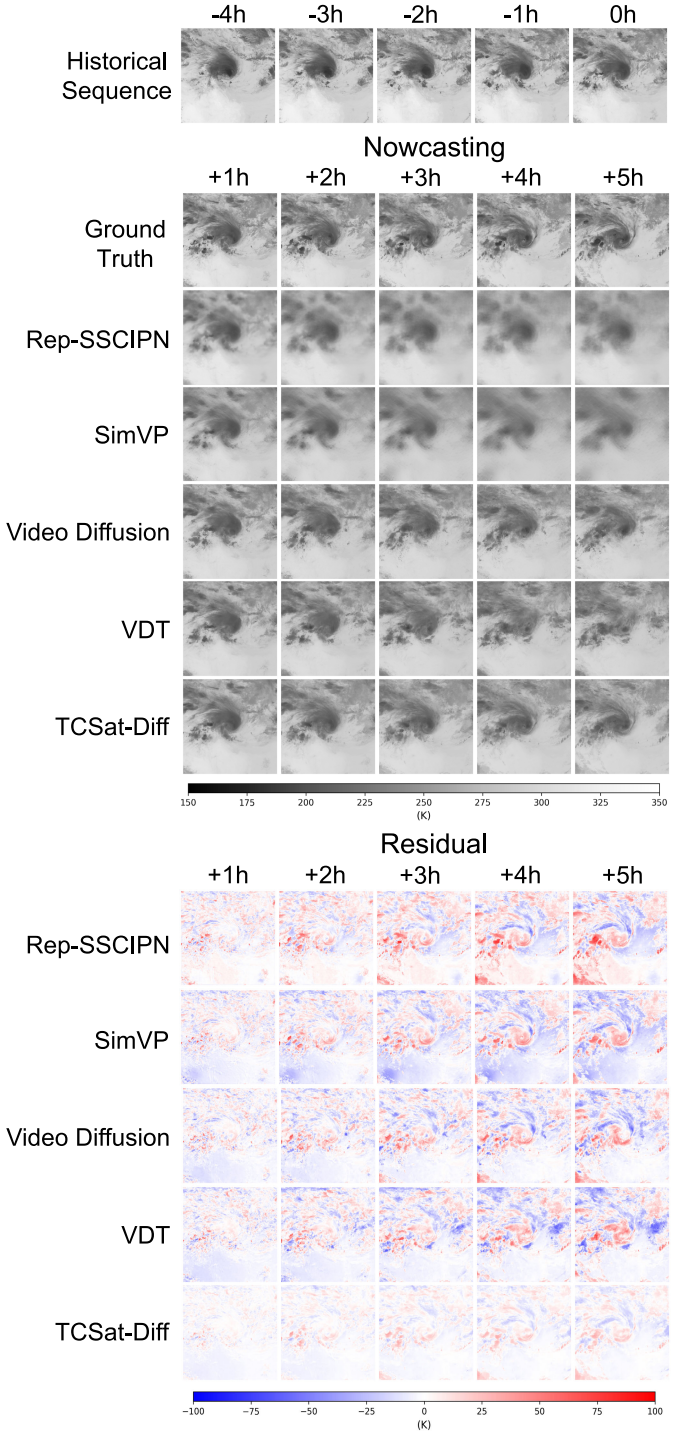


Fig. 8. Visual analysis of TC Kirrily (2024) in the Southern Hemisphere, using TCSCI inputs from 03:00 to 07:00 UTC on January 25, 2024, to predict the evolution from 08:00 to 12:00 UTC. The results demonstrate that TCSat-Diff achieves high visual fidelity when applied to intense TC in the previously unseen Southern Hemisphere, underscoring its strong cross-regional generalization capability.

attribute this phenomenon to the smoothing effect inherent in ensemble averaging. As the ensemble size increases, the averaging operation suppresses stochastic and high-frequency variations, producing more deterministic and smoother outputs. This smoothing effect improves statistical consistency but inevitably reduces fine-grained details and perceptual

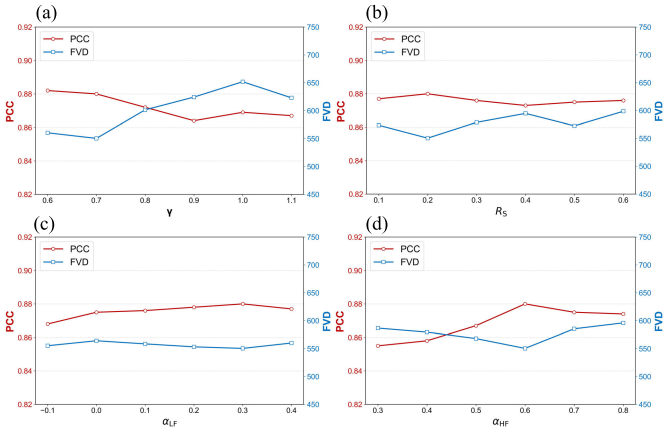


Fig. 9. Sensitivity analysis of the NFM module to four modulation parameters. (a) γ, controlling the initial strength ratio of high-frequency components. (b) R_S, denoting the coefficient of the maximum distance from the frequency center that defines the mask boundary. (c) α_{LF}, determining the enhancement strength of low-frequency modulation. (d) α_{HF}, governing the enhancement strength of high-frequency modulation. PCC (red) and FVD (blue) evaluate statistical accuracy and fidelity, respectively.

diversity in the generated results. In cross-regional generalization tests on 480 South Pacific TCs, TCSat-Diff maintains excellent accuracy and visual quality despite the substantial dynamic differences in TC behavior, demonstrating robust cross-regional generalization without additional training.

However, future work can be extended to improve the prediction quality. First, the efficiency evaluation indicates that the diffusion-based sampling process introduces high computational costs. Nevertheless, adopting more efficient samplers demonstrated promising reductions in inference time with minimal performance loss in our experiments. Future work will explore more advanced samplers to improve modeling efficiency. Second, the proposed model currently employs only TCSCI as input. Its performance can be further enhanced by incorporating multimodal data (e.g., multichannel satellite observations, radar retrievals, and numerical simulations), since the evolution of cloud systems is strongly modulated by environmental variables. In addition, we intend to embed additional physical priors into TCSat-Diff, such as constraints on energy conservation and potential vorticity, to further improve the model's generalization capability and enhance its physical consistency and interpretability.

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